IGORS VOLOHOVS

C/C++ Developer

CONTACTS

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LINKS

t.me/IgorsVolohovs

linkedin.com/in/igors-volohovs-764850221/

github.com/IGORSVOLOHOVS

SKILLS

- C++23
- Python 3.13
- Qt Framework
- Windows API
- Visual Studio
- CMake
- Github & Actions
- SWIG
- Google Antigravity
- Linux
- Conan
- FreeRTOS
- UML 2.0 / C4 Model
- Jira / Confluence
- Rust / Java / C# / Julia
- OpenGL
- STM32CubeIDE
- GTest / GBenchmark
- Docker / n8n
- Jenkins
- SQL

LANGUAGES

English	Professional
Latvian	Native
Russian	Native

HOBBIES

- Assemble a 3×3 - 7×7 Rubik's Cube
- Learning Maths & Developing my own "Theory of effective thinking and task solving"

ABOUT ME

C++ Developer focused on solving complex technical challenges and low-level system optimization. I am a proactive learner who enjoys deep-diving into technology and sharing knowledge with the team.

WORK EXPERIENCE

Software Developer in TSI AdditiveLab
Jan 2024 - Present

TSI | Riga, Latvia (Wire Arc Additive Manufacturing)

Developing **WaamMonitoring** — a comprehensive software for real-time monitoring, graphical analysis, and data storage of the metal 3D printing process (my Bachelor's thesis project). Implemented **WaamAPI**: a low-level HAL in C++23 with Python 3.13 bindings for seamless hardware-software integration.

Hardware: Yaskawa GA-50, Yaskawa HC-10, Wenglor MLZL171 (2D profilometer), Optris CTi Pyrometers, Fronius TPS 500i.

Technologies used: C++23, Python 3.13, Qt Framework, Windows API.

Software Developer (Edge AI)
Jan 2026 - Present

TSI AdditiveLab | Riga, Latvia

Edge AI device based on Rockchip RV1106. Preparation of dataset, training and deployment of YOLOv8n neural network for defects localization. The goal of this project is to create a low-cost Edge AI device (robot mounted) to control quality of welding seems in AdditiveLab.

Technologies used: YOLOv8n, Python, C++, OpenCV, SSH, RKNN-Toolkit2.

Medical Device Embedded Firmware Developer
Dec 2022 - Present

IDEAHUB | Riga, Latvia

Development of firmware for a medical cell staining machine. Connected various sensors, motors and network devices to STM32F4 MCUs, and wrote a driver for UART-USB module to connect Raspberry Pi with several STM32F4 MCUs.

Technologies used: C/C++11, STM32CubeIDE, Linux, Rust, SWIG.

ESF Project Developer
Feb 2023 - Jul 2023

TSI | Riga, Latvia

Creating software for the mobile robot KOALA. The goal was to create a user-friendly development environment in C/C++ and Python for creating new projects.

Technologies used: CMake, C/C++11, Python, Linux.

Private Tutor (Math & Programming)
Sep 2022 - Present

Teaching Mathematics and Programming to schoolchildren, helping them build strong logical foundations and practical coding skills.

EDUCATION

Computer Vision Course
Present

Yandex Practicum

2 projects completed; working with OpenMMLab, PyTorch, ClearML, CVAT.

C++ Middle Developer
2025

Yandex Practicum

9 projects; advanced C++20/23 and introduction to C++26 standards.

Computer Science Bachelor Program
2021 - Present

TSI, Riga

C++ Developer (~300 tasks were solved)
2021

Yandex Practicum

Basic Course for Network Management
2019

ITLAT, Riga

Basic Course of Java
2018

ITLAT, Riga

Basic Course of C/C++
2017

ITLAT, Riga